Arjan Bista

Undergraduate 4th year Mechanical Engineering Student
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Profile

Undergraduate Mechanical engineering student at IOE Pulchowk Campus, Tribhuvan University, with hands-on experience in robotics, mechanical design, fabrications and automation through academic project, national and international undergraduate robotics competition. Proficient in Solidworks, Ansys, Pybullet, ROS,Python,C/C++ and practical manufacturing process. Involved in developing, fabricating, testing and improving robotic system and testing with additional exposure to Reinforcement Learning (RL), Large Language Models (LLMs), and ROSA.

Work Experiences

Solidworks CAD Designer(Contract based)

Smart Cheli-Nepal
Dec 2025-Jan 2026

- Designed 3D parts for CNC laser cutting and 3D printing

AI Intern

Accelerated Komputing(Ank)-Nepal
Oct 2025-Nov 2025

- Worked on Rosa and LLM based language command on ros2 to control the robot.
- Developed Rosa and LLM integrated four-wheel robot in Gazebo in ROS which was controlled by natural language processing.

Mechanical CAD Designer

Nagarkot Zip Coaster-Nepal
Jul 2025-Sep 2025

- Designed and developed zip liner roller assembly.
- Created detailed CAD models, assemblies and manufacturing drawings for fabrication.
- Refined the design for compatibility, manufacturability, functionality and ease of assembly.

Education

SEE (Grade 10) – Lalitpur Madhyamik Vidyalaya (LMV)

Completed:2020 | GPA: 3.90

Grade 11 & 12 (Science) – Sainik Awasiya Mahavidyalaya, Bhaktapur(SAMB)

Completed:2022 | GPA:3.58

Bachelor of Engineering in Mechanical Engineering – IOE, Pulchowk Campus, Tribhuvan University

Currently 4th year– Expected graduation:2027 | Percentage: 70.21%(Aggregate up to 6 sem)

Technical Skills

Programming Languages: Python, C/C++,Javascript

CAD Tools:	Solidworks, AutoCAD
Simulation Tools & Robotics:	Ansys, MATLAB ,Gazebo,Pybullet,ROS2,SLAM
Artificial intelligence & Machine Learning:	Large Language Models(LLMs), Reinforcement Learning(RL), ROSA
Hardware and Electronics:	Arduino,Motordrivers,Rasberry Pi, RFID, Distance/ Proximity sensor, Servo Motors, DC , Stepper Motor ,Bldc Motor,Color Sensors
Development Tools:	Git,Github, Vscode
Manufacturing & Fabrication:	Milling, Lathe, Welding,3d printing, grinding, CNC laser cutting
Engineering & System Development:	Mechanical Design, Robot Design, CAD Modeling, Fabrication, Assembly, Testing, System Integration

Project Competitions & Achievements

- **STEAM Project 2026 -Clay 3D printer:** Successfully converted FDM 3D printer to clay 3d printer. Focusing in nozzle design, pump design, power transmission and printer control and clay behavior. Performed advance modeling, manufacturing, assembly, testing, design improvement while coordination with team ensuring timely completion.
- **SIC Project -Robotic Arm Manipulation using Reinforcement Learning:** Designed and build robotic arm in SolidWorks and converted the CAD into URDF for simulation. Implemented the URDF model in PyBullet and trained a Reinforcement learning (RL) with Proximal Policy Optimization (PPO)) for robotic manipulation. Trained robot could reach and manipulate target position during testing.
- **KU hackfest 2025 - Semantic Navigation for TurtleBot3 with LLM Integration:** Implemented Navigation2(Nav2) for autonomous movement across multiple rooms. Integrated ROSA for intelligent control. Developed an OpenCV vision pipeline to enhance image clarity. Integrated GPT-4o for context-aware identification of geometric shapes.
- **ABU Robocon 2025 - Basketball:** Built robots for dribbling, passing, and shooting. Led mechanical design, advanced CAD modeling, manufacturing, and assembly. Implemented design improvements, coordinated the team, and ensured timely completion. Strengthened leadership, management, and technical skills.
- **IMAC BE Design Competition by SOMAES:** Worked on the theme of rescue, where we developed an autonomous rescue vehicle, a water-surfer for rescue. The project won the Technical Award.
- **ABU Robocon 2024 -Harvest Day:** Built robots for planting seedlings, throwing balls, and collecting balls autonomously. Gained hands on experience in design, development, testing, and team coordination. Won the Paramount Award.
- **MechTRIX 2081- BE Design Hackathon by SOMAES:** Worked on a modular, versatile, and frugal small-scale agricultural machine for cash crop cultivation. Designed different modular systems for planting various crops.
- **ABU Robocon 2023 - Casting Flowers Over Angkor Wat:** Built two robots for ring picking and throwing. Learned basic machining, CAD modeling in SolidWorks, and part fabrication. Contributed to assembly and testing. Won Mabuchi Award.
- **MechTRIX 2080-BE Design Hackathon by SOMAES:** Worked with my team on the theme of automation to design a basic automatic college gate system. The design was simple due to limited CAD practice, but we successfully implemented the core concept.
- **School level competition:** Actively participated in annual school-level ICT and Science Fairs since Class 8, presenting projects such as an Automatic Door Lock System, an RFID-based Lock System, and an Arduino-powered Obstacle Avoiding Car, showcasing a strong interest in robotics and automation.

Workshops and Training

- **Solidwork Fellowship -Robotics Club (2023)**: Workshop on covering fundamental uses of CAD software.
- **RoboCamp- Robotics Club (2023):** Hands on workshop covering Raspberry Pi, motordrivers and robotics fundamental
- **SIC (Samsung Innovation Campus) -Programming Training:** Completed a programming focused training program at IOE, Pulchowk Campus**,** covering Python programming, programming logic, data structures, and algorithms**.**